

A Practical Introduction to Machine Learning in Python

Day 4 – Thursday

»Supervised Machine Learning«

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September 19, 2024

Today

Cross-validation

Finding the optimal (hyper-)parameters

Grid search

A typical case for gridsearch: The regularization parameter C

More suggestions for improving your models

Tuning decision thresholds with ROC curves

ML versus dictionaries

The explainability of ML

Exercise

Cross-validation
oooooooo

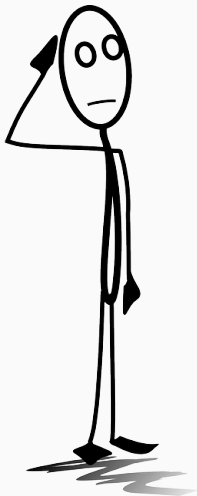
Finding the optimal (hyper-)parameters
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More suggestions
oooooooo

ML versus dictionaries
oooooooo

Exercise
ooo

Also read chapter 8.5!



Everything clear from this morning?

Cross-validation

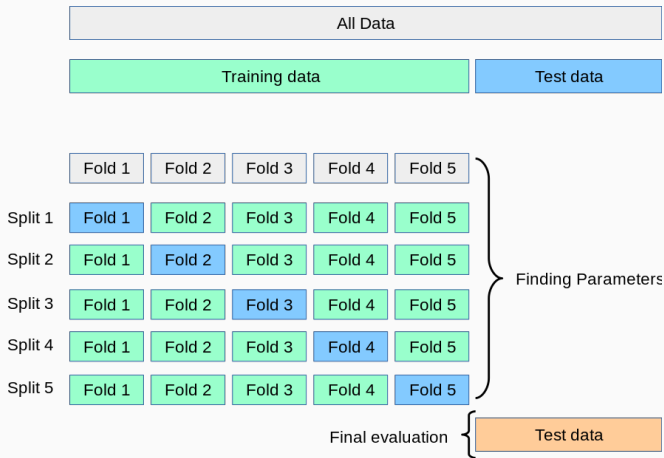


Figure 1: First, we set aside a test dataset for the final evaluation. Then, we split our data into $k = 5$ folds, where each fold is used for validation once (blue) and $k - 1 = 4$ times for training. Source: https://scikit-learn.org/stable/_images/grid_search_cross_validation.png

Cross-validation

```
1 from sklearn.model_selection import cross_val_score
2 from sklearn.naive_bayes import MultinomialNB
3 nb = MultinomialNB() # the classifier we trained this week
4 scores = cross_val_score(nb, train_features, [r[1] for r in reviews], cv
    =10)
5 print(scores)
```

results in:

```
1 [0.858 0.8612 0.8516 0.8528 0.8672 0.8664 0.8576 0.8652 0.8436 0.852 ]
```

We estimate the model 10 times on different training/validation data splits and get 10 different evaluation scores (here: accuracy, but we can use precision, recall, F1, etc. – see examples in the book).

Note that a simple 50:50 train/validation split is identical to setting $k = 2$

Reasons to do Cross-validation

We can get confidence intervals around the scores

- If we have 10 scores instead of one, we can not only get the mean, but also a standard deviation
- If you have two models, both with a mean accuracy (or F1, or whatever) of .85 but one with a large and one with a low standard deviation, you probably prefer the latter – it generalizes better, less likely to suffer from overfitting

Reasons to do Cross-validation

We do not “waste” too much validation data

- If $k = 10$, in each iteration, we use 90% of the data for training.
- We even use *all* data (except test set, of course) for training at least once.
- With train/validation split, we probably need a larger validation set to be sure (e.g., 80/20)
- Especially relevant when annotation is expensive.

Takeaway i

Simple train/test split

- introductory examples/pedagogical reasons
- really small dataset where you cannot afford to set aside validation data
- if – for whatever reason – you do not compare different models and settings.

Takeaway ii

Train/validation/test split

- To compare different model configurations without overfitting on the test data
- Pedagogical reasons, simple start
- Very large datasets or very complex models where setting $k > 2$ would lead to prohibitively expensive computations

Takeaway iii

k-fold crossvalidation

- Best option for comparing multiple models, *especially* when engaging in *hyperparameter tuning* (next section)

In other words: whether you have or create
a train/validation/test or use
cross-validation – just make sure you always
set aside one test set for final evaluation
that you have never used before.

Finding the optimal (hyper-)parameters

hyperparameter a parameter of a model that is not learned through training, but specified in advance

Finding the optimal (hyper-)parameters

Grid search

Hyperparameter optimization with grid search

General idea

Rather than arbitrarily trying some configurations, let's systematically test and compare.

First idea: Use a for-loop! (Example 11.4 in the book)

```
1 configurations =
2 [('NB with Count', CountVectorizer(min_df=5, max_df=.5), MultinomialNB()
3   ),
4   ('NB with TfIdf', TfidfVectorizer(min_df=5, max_df=.5), MultinomialNB())
5   ,
6   ('LogReg with Count', CountVectorizer(min_df=5, max_df=.5),
7     LogisticRegression(solver='liblinear')),
8   ('LogReg with TfIdf', TfidfVectorizer(min_df=5, max_df=.5),
9     LogisticRegression(solver='liblinear'))]
10
11 for description, vectorizer, classifier in configurations:
12     print(description)
13     X_train = vectorizer.fit_transform(reviews_train)
14     X_test = vectorizer.transform(reviews_test)
15     classifier.fit(X_train, y_train)
16     y_pred = classifier.predict(X_test)
17     short_classification_report(y_test, y_pred)
18     print('\n')
```

(where (X_test, y_test) hopefully refers to a validation dataset with another test dataset set aside)

A wishlist

We now tested 2×2 combinations: (NB|LogReg) $\times$ (count|tf·idf).

Wouldn't it be nice if we...

- could simply state this instead of manually creating the list?
- (especially if we had (NB|LogReg) $\times$ (count|tf·idf) $\times$ (min_df=5|min_df=1) $\times$ (max_df=.5|max_df=.8|max_df=.9) = $2 \times 2 \times 2 \times 3 = 24$ options)
- would not have to check manually which one performed best?

That's what scikit-learn's `gridsearch` functionality does! (and the second bullet point is called the *grid*)

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That's what scikit-learn's gridsearch functionality does! (and the second bullet point is called the *grid*)

Gridsearch is especially useful for **hyperparameter optimization**, such as trying out different values for `min_df` and `max_df` on the previous slide.

You could also use a combined approach where you first use a handwritten loop to narrow down the number of candidate models and then tune the model(s) you settled on with a grid search.

Finding the optimal (hyper-)parameters

A typical case for gridsearch: The regularization parameter C

The regularization parameter λ (or $C = .5n\lambda$)

$$\arg \min_{\beta} \left[\sum_{i=1}^n \left(Y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ji} \right)^2 + \lambda \sum_{j=1}^p |\beta_j|^q \right]$$

We estimate the β -coefficients of a model by optimizing a so-called loss function, i.e minimizing the “cost” that occurs when the prediction is wrong.

If we simply add the sum of the β coefficients to the model, we “punish” large coefficients.

If λ is larger, we punish more.

If $q = 1$, we call it L1 regularization. If we add the squared coefficients ($q = 2$), we call it L2 regularization.

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- scikit-learn uses regularization by default
- (that's why the coefficients of a logistic regression classifier will differ from a "traditional" logistic regression with the statsmodels module)
- Remember that we are *not* interested in a substantive interpretation of the β -coefficients but in a good prediction (which is why "punishing" large coefficients and hence biasing the coefficients is not a problem at all!)
- The penalty will shrink the coefficients towards zero, *which is very useful to avoid overfitting!*
- As always, there is a lot of extra info on the scikit-learn website

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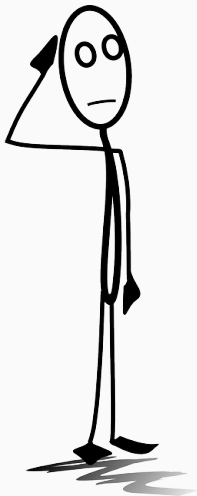
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How harsh should the penalty be?

Hyperparameter optimization with grid search

General idea

Rather than arbitrarily trying some combinations of hyperparameters, let's systematically test and compare.

Example

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- To avoid overfitting, scikit-learn adds a *regularization term* to the loss function that is minimized to fit the regression.
- Think of this term as a penalty for overfitting
- How much weight should our penalty carry? That's determined by a constant, C .
- How to determine the best C ? $\Rightarrow$ grid search

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Hyperparameter optimization with grid search

Finding C in a logistic regression using 5-fold cross-validation

```
1 from sklearn.linear_model import LogisticRegressionCV
2 logregCV = LogisticRegressionCV(cv=5).fit(train_features, [r[1] for r in
    reviews])
```

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```

1 pipeline = Pipeline(steps = [('vectorizer', TfidfVectorizer()), ('
    classifier', LogisticRegression(solver='liblinear'))])
2 grid = {
3     'vectorizer__ngram_range' : [(1,1), (1,2)],
4     'vectorizer__max_df': [0.5, 1.0],
5     'vectorizer__min_df': [0, 5],
6     'classifier__C': [0.01, 1, 100]
7 }
8
9 search = GridSearchCV(estimator=pipeline,
10                       param_grid=grid,
11                       scoring='accuracy',
12                       cv=5,
13                       n_jobs=-1, # use all cpu's
14                       verbose=10)
15 search.fit(reviews_train, y_train)
16 print(f'Using these hyperparameters {search.best_params_}, we get the
    best performance:')
17 print(short_classification_report(y_test, search.predict(reviews_test)))

```

Example 11.6 (book)

Note that the grid is specified as a dictionary where the keys are strings with the name of the step in the pipeline, followed by two underscores, followed by the parameter to tune, and the values are lists of values.

This means that *any* parameter that the classifier or vectorizer takes can be tuned!

The pipeline notation in scikit learn

You might have noticed the Pipeline construct in the last example.

- Machine learning involves multiple steps (e.g., preprocessing → vectorizer → classification)
- We did all of them separately before
- Nothing wrong with that, but to ease use and evaluation of *the whole process* (as needed for the gridsearch), we can define a pipeline.

Example with a pipeline (and add cross-validation)

```
1 from sklearn.feature_extraction.text import TfidfVectorizer
2 from sklearn.linear_model import LogisticRegressionCV
3 from sklearn.pipeline import make_pipeline
4
5 vec = TfidfVectorizer()
6 clf = LogisticRegressionCV()
7 pipe = make_pipeline(vec, clf)
8
9 pipe.fit([r[0] for r in reviews], [r[1] for r in reviews])
10 predictions = pipe.predict([r[0] for r in test])
```

Pipeline takeaway

- In principle, just a different way to write what we already did
- The more steps, the more relevant (e.g., preprocessing → vectorizer → dimensionality-reduction → classification)
- The more you rely on automated evaluation (e.g., grid search) of *multiple* steps in the pipeline, the more useful it is

Grid-search takeaway

- When you want to systematically test what happens when you vary a hyperparameter, use grid-search to automatically do so and select the best value
- sometimes already implemented (e.g.,
LogisticRegressionCV as direct replacement for
LogisticRegression)
- But GridSearchCV is very flexible: can be used in
combination with pipeline for very different purposes

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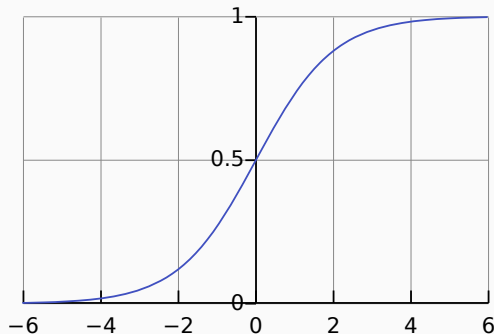
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More suggestions

More suggestions

Tuning decision thresholds with ROC curves

From estimate to label



In logistic regression, we use the *sigmoid function* to transform the estimates into probabilities.

To transform the probabilities into binary labels, we use a cutoff (default: 0.5).

Why use 0.5 as cutoff?

- It makes most sense (intuitively, mathematically)
- But remember our precision/recall tradeoff: maybe we want to be 'stricter' or 'less strict'
- Maybe it is important to us that our classifier is balanced and equally good in predicting both classes, even if overall accuracy suffers (slightly)

Let's see what happens if we plot False Positives against True Positives (ROC-curve)

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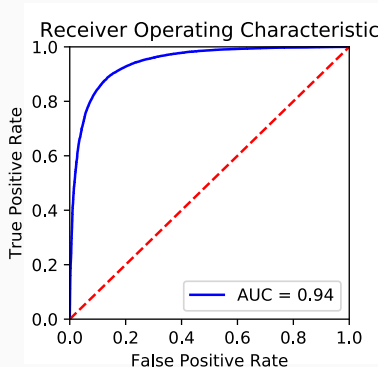
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ROC Curve



- If we choose a threshold such that we get very little false positives, we also get too little true positives.
- Optimum in the upper left corner

Some further ideas to look into

Balancing classes

Your classifier probably works better if you have approximately the same amount of annotated training data for both classes (e.g., pos/neg). If getting such data is not an option, you may consider weighing accordingly, e.g. using

```
LogisticRegression(class_weight='balanced')
```

Some further ideas to look into

More advanced pipelines

Consider constructing advanced pipelines, including a dimension reduction step:

https://scikit-learn.org/stable/auto_examples/compose/plot_digits_pipe.html

Some further ideas to look into

Combine different feature sets

E.g, use BOW-features as well as features such as sentence length, number of sentences (or whatever)

<https://scikit-learn.org/stable/modules/generated/sklearn.compose.ColumnTransformer.html>

ML versus dictionaries

What *is* our fitted classifier again?

Essentially, just a formula

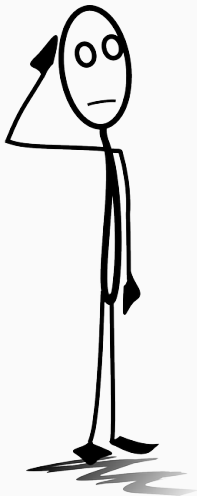
$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}}$$

where β_0 is an intercept¹, β_1 a coefficient for the frequency (or tf·idf score) of some word, β_2 a coefficient some other word.

If our fitted *vectorizer* contains 5,000 words, we thus have 5,001 coefficients.

(for logistic regression in this case, but same argument applies to other classifiers as well)

¹Machine Learning people sometimes call the intercept “bias” (yes, I know, that’s confusing)



But isn't that then essentially very much like a dictionary, except that the words have different weights?

In some sense, yes.

- But we don't pretend that we can construct the dictionary *a priori*.
- It's specifically tailored to our use case.
- The weights are *really* essential here.

We *could* print all coefficient-word pairs, but probably it's enough to just look at those with the largest absolute value:

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ML versus dictionaries

The explainability of ML

ELI5

```
In [98]: import eli5
eli5.show_weights(pipe, top=10)
```

```
Out[98]: y=1 top features
```

Weight?	Feature
+9.043	great
+8.487	excellent
+6.908	perfect
... 37662 more positive ...	
... 37178 more negative ...	
-6.507	worse
-7.347	poor
-8.341	boring
-8.944	waste
-8.976	bad
-9.152	awful
-12.749	worst

```
In [111]: eli5.show_prediction(clf, test[0][0],vec=vec)
```

```
Out[111]: y=1 (probability 0.844, score 1.689) top features
```

Contribution?	Feature
+1.920	Highlighted in text (sum)
-0.232	<BIAS>

it is a rare and fine spectacle, an allegory of death and transfiguration that is neither preachy nor mawkish. a work of mature and courageous insight, northfork avoids arthouse distinction by refusing to belong to a kind. unlike the most memorable and accomplished film to impose an obvious comparison, wim wenders' 1987 wings of desire (der himmel über berlin), it sustains an ambivalence in a narrative spectrum spanning from the mundane to the supernatural. this story of earthly and celestial eminent domains in the american west withholds the fairytale literalness that marked its german predecessor in the ad hoc genre of angels shedding their wings with obsequious sentimentalism. its celestial transcendence, be it inspired by doleful faith or impelled by a fever dream, never parts ways with crud and rot. this firm grounding redounds to great credit for writers and directors mark and michael polish.

ELI5

- Inspecting *all* coefficients of a ML model usually doesn't make much sense
- But that does not mean that we cannot understand how the model makes its predictions
- We can look at the most important coefficients
- We can look which words in a given text contributed most to its classification

Have we solved all problems of dictionaries?

No.

For instance, the negation and/or intensifier problem.

Possible approaches

- n -grams as features
- preprocessing (?)
- deep learning
- ...

⇒ But ultimately, it's just an empirical question how big the problem is!

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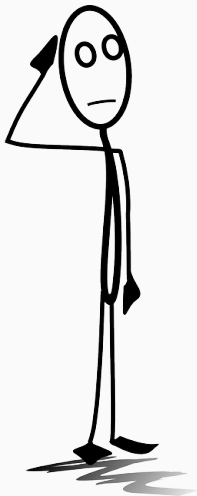
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Exercise



Any questions?

Cross-validation
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Finding the optimal (hyper-)parameters
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More suggestions
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ML versus dictionaries
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Exercise
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Let's exercise!